STA 35C: Statistical Data Science III

Lecture 8: Classification Basics& Logistic Regression

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Spring 2026, UC Davis

Announcements

Homework 3 is posted (Due: Tue, Apr 21, 11:59 PM)

- Please submit on time and follow the submission instructions
- Please review the homework problems early so you have time to ask questions if needed
- Feel free to ask questions during lecture, in office hours or post them on Piazza

Midterm 1 is in class on Fri, Apr 24

- You may bring *one **hand-written** letter-sized (8.5 $\times$ 11 inches), double-sided sheet of paper* with formulas and brief notes
- **Calculator**: Simple (non-graphing) calculators only
- **No textbooks** or other materials beyond the single cheat sheet
- **SDC accommodations:** Confirm scheduling with AES online

I'll be holding my office hours today 3:30–4:30 PM at MSB 4220

- Bring your questions

Agenda

So far:

- Regression as a supervised learning problem
- Linear regression for quantitative responses

Today:

- Classification: what it is and how it differs from regression
- Logistic regression for binary responses
  - Basic ideas
  - Model formulation
  - Prediction and interpretation
  - Parameter estimation

Classification: Motivation

Classification = Supervised learning to predict *qualitative* (categorical) responses

Examples:

- Email spam vs. non-spam
- Fraudulent transaction vs. legitimate
- Medical diagnosis (multiple possible conditions)
- Handwritten digit classification (0–9)

Key differences from regression:

- *Response:* Y is a *class label*, not a numeric value
- *Prediction:* Many methods estimate class probabilities, then convert them into class labels
- *Assessment:* Performance is evaluated differently (e.g., error rate, confusion matrix)

Classification: A visual illustration

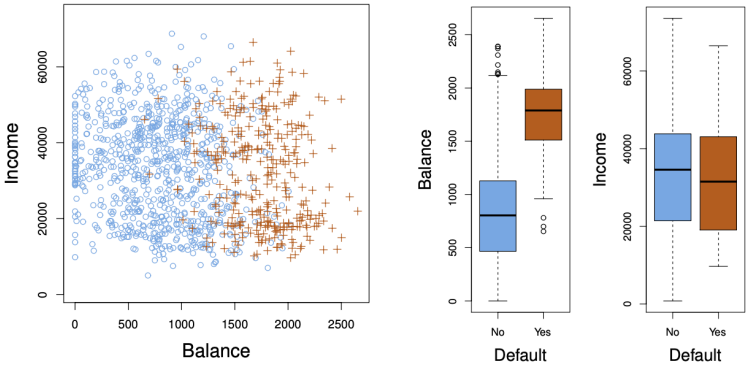


Figure: The `Default` dataset: annual incomes vs. monthly credit card balances. Orange: individuals who defaulted, Blue: those who did not [JWHT21, Figure 4.1].

Goal: Learn a decision rule (or decision boundary) that assigns a new point x_{new} to a class

Classification setting: Formal description

Goal: Given labeled data $\{(x_i, y_i)\}_{i=1}^n$, learn a rule that assigns x_{new} to one of K classes

Examples:

- Email text $(X) \to$ spam or not (Y)
- Handwritten image $(X) \to$ digit (Y)
- Patient measurements $(X) \to$ medical condition (Y)

Typical procedure:

1. Estimate class probabilities:
$$p_k(x) = \Pr(Y = k \mid X = x), \qquad k = 1, \ldots, K,$$
2. Predict the class with the largest estimated probability:
$$\hat{y}(x) = \arg\max_{k \in \{1,\ldots,K\}} \hat{p}_k(x)$$

Question: Wait... why not just use regression methods?

Why don't we simply use linear regression?

Naive attempt 1: Treat classes as numbers

- Encode $Y \in \{0, 1\}$ or $\{1, \ldots, K\}$ numerically
- Fit a linear model $Y \approx \beta_0 + \beta_1 X$

$\rightarrow$ For $K > 2$, coding classes as $1, 2, \ldots, K$ imposes an artificial ordering and spacing

$\rightarrow$ The fitted value need not be one of the class labels

Naive attempt 2: Model the class probability linearly

- For binary $Y \in \{0, 1\}$, let $p(x) := \mathbb{E}[Y \mid X = x] = \Pr(Y = 1 \mid X = x)$
- Try a linear model $p(x) \approx \beta_0 + \beta_1 x$

$\rightarrow$ Linear regression can produce fitted values outside $[0, 1]$, which are not valid probabilities

$\rightarrow$ For binary data, $\mathrm{Var}(Y \mid X = x) = p(x)(1 - p(x))$, so the constant-variance assumption of linear regression is violated

Takeaway: We need a method that respects the *categorical* nature of Y and keeps predicted probabilities in $[0, 1]$

Visual illustration: Linear vs. logistic regression

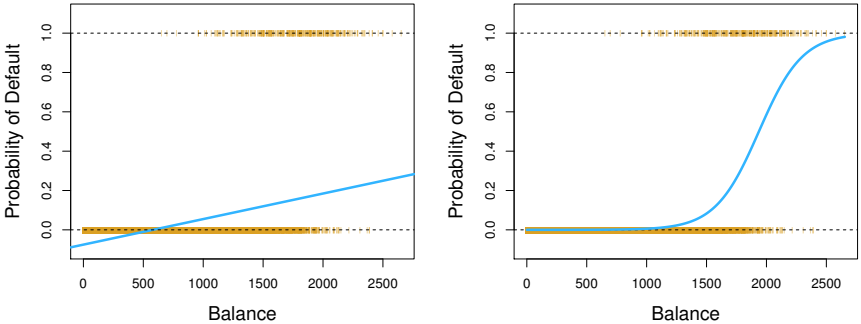


Figure: **Left:** Estimated "probability" of `default` using linear regression. **Right:** Probability estimated via logistic regression [JWHT21, Figure 4.2].

Therefore: Classification-specific methods are typically more appropriate and preferred

Pop-up quiz: Classification vs. regression

Suppose we code a binary response as $Y \in \{0, 1\}$ and fit a linear regression model. For some predictor value x, the fitted value is $\hat{y} = -0.15$.

Question: What is the best conclusion?

A) The estimated probability of class 1 is -15%.

B) Linear regression can produce fitted values outside $[0, 1]$, so it is not ideal for modeling class probabilities.

C) The model predicts class 0 with certainty.

D) The data must contain mislabeled observations.

Answer: B. A fitted value of -0.15 cannot be a valid probability, which is one reason we prefer classification-specific methods.

Roadmap

We will learn two types of classification methods

- **Logistic regression**: a *discriminative* approach that models $P(Y = 1|X)$
- **Generative models**: model $P(Y)$ and $P(X \mid Y)$, then apply Bayes' rule
  - Example: Linear discriminant analysis (LDA), Naive Bayes

Today: Logistic regression with one predictor X ($p = 1$) for binary ($K = 2$) classification

Next:

- Fri, Apr 17: Logistic regression (extensions) & classification errors
- Mon, Apr 20: Linear discriminant analysis
- Wed, Apr 22: Linear discriminant analysis (cont'd)
- Fri, Apr 24: Midterm 1

Logistic regression: From probabilities to log-odds

Binary classification: For binary response ($Y \in \{0, 1\}$), let

$$p(x) = \Pr(Y = 1 \mid X = x).$$

- We want to model $p(x) \in [0, 1]$.
- Then predict the class using the estimated probability $\hat{p}(x)$:

$$\widehat{Y} = \begin{cases} 1, & \text{if } \hat{p}(x) \geq p^* \quad (\text{often } p^* = 0.5), \\ 0, & \text{otherwise.} \end{cases}$$

Question: How can we connect probabilities to a linear model?

- A linear model for $p(x)$ is problematic because probabilities must stay in $[0, 1]$
- So we look for a transformation $p \mapsto g(p)$ that is
  - monotone, so larger p means stronger evidence for class 1;
  - able to take any real value, so we can model $g(p(x))$ linearly
- A natural choice is: probability $\to$ the odds $\to$ the log-odds (logit)

$$p(x) \qquad \to \qquad \frac{p(x)}{1 - p(x)} \in (0, \infty) \qquad \to \qquad \log\left(\frac{p(x)}{1 - p(x)}\right) \in \mathbb{R}$$

Logistic regression: Model formulation

The logit is especially convenient because it is monotone, maps $(0, 1)$ to $\mathbb{R}$, and turns multiplicative changes in odds into additive changes

Key idea: Model the log-odds as a linear function of x:

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$$

- Positive log-odds $\iff p(x) > 0.5$
- Negative log-odds $\iff p(x) < 0.5$

Question: How do we write $p(X)$ as a function of β_0, β_1, X?

- Solving for $p(x)$,

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x \quad \iff \quad \frac{p(x)}{1-p(x)} = e^{\beta_0+\beta_1 x}$$

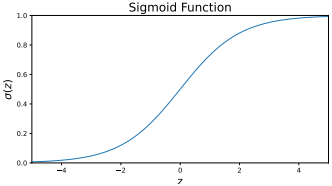
$$\iff \quad p(X) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}} = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}}$$

Logistic regression model

Logistic regression model:

$$p(x) = \Pr(Y = 1 \mid X = x) = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)} =: \sigma(\beta_0 + \beta_1 x)$$

- $\sigma(z) = \frac{e^z}{1+e^z}$ is the **logistic** (sigmoid) function
- $p(x) \in (0, 1)$ for all x
- If $\beta_1 > 0$, then larger x means larger odds and a larger class-1 probability; if $\beta_1 < 0$, the opposite holds
- A one-unit increase in x changes the *log-odds* by β_1, so the *odds* are multiplied by e^{β_1}



Class prediction with logistic regression model

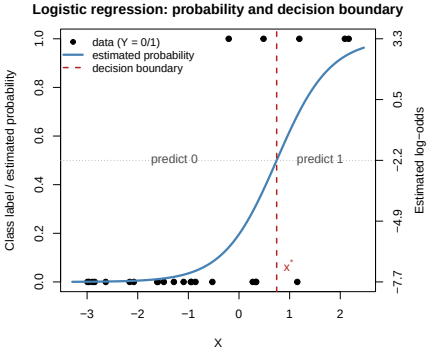
Logistic regression model: $p(x) = \Pr(Y = 1 \mid X = x) = \frac{\exp(\beta_0+\beta_1 x)}{1+\exp(\beta_0+\beta_1 x)}$

Decision rule:

$$\widehat{Y} = 1 \qquad \Longleftrightarrow \qquad \hat{p}(x) \geq p^* \qquad \Longleftrightarrow \qquad \hat{\beta}_0 + \hat{\beta}_1 x \geq \log\left(\frac{p^*}{1-p^*}\right)$$

- If $p^* = 0.5$, then $\log\left(\frac{p^*}{1-p^*}\right) = 0$, so predict class 1 when $\hat{\beta}_0 + \hat{\beta}_1 x \geq 0$
- For $p^* = 0.5$, the decision boundary is the point where $\hat{\beta}_0 + \hat{\beta}_1 x = 0$, i.e. $x^* = -\hat{\beta}_0/\hat{\beta}_1$ when $\hat{\beta}_1 \neq 0$

Question: What if we increase p^* from 0.5 to 0.9?



An example in R: Fraud or not?

Scenario: Predict if a transaction is fraud ($Y = 1$) or not ($Y = 0$) using its amount (X)

```
# Simulate toy data:
set.seed(123)
n <- 100
X <- runif(n, 1, 500)  # transaction amount
# true logistic function: p = 1 / [1 + exp(-(-5 + 0.02*X))]
p <- 1 / (1 + exp(-(-5 + 0.02*X)))
Y <- rbinom(n, 1, prob=p)

# Fit logistic regression:
model <- glm(Y ~ X, family=binomial)
summary(model)

# Probability of fraud at X=300:
predict(model, data.frame(X=300), type="response")
```

```
Call:
glm(formula = Y ~ X, family = binomial)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -6.084377   1.242746  -4.896 9.79e-07 ***
X            0.024664   0.004923   5.010 5.44e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 138.629  on 99  degrees of freedom
Residual deviance:  53.504  on 98  degrees of freedom
AIC: 57.504

Number of Fisher Scoring iterations: 6

> # Probability of fraud at X=300:
> predict(model, data.frame(X=300), type="response")
        1
0.7883033
```

Check:

- Interpret $\hat{\beta}_0, \hat{\beta}_1$
- $\exp(\hat{\beta}_1)$: multiplicative change in the odds for a \$1 increase in amount
- `type="response"` returns $\hat{p}(x)$, not the log-odds

Pop-up quiz: Logistic regression coefficients

Suppose you fit the model

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x, \qquad p(x) = \Pr(Y = 1 \mid X = x)$$

and obtain $\hat{\beta}_1 = -2.0$.

Question: Which interpretation is most accurate?

A) The estimate is invalid because logistic regression requires $\beta_1 > 0$.
B) For each one-unit increase in X, the probability $p(x)$ decreases by exactly 2 percentage points.
C) For each one-unit increase in X, the odds of $Y = 1$ are multiplied by $e^{-2} \approx 0.14$.
D) If X increases enough, the predicted probability becomes negative.

Answer: C. A one-unit increase in X decreases the log-odds by 2, so the odds are multiplied by $e^{-2} \approx 0.14$; the probability changes nonlinearly, not by a fixed amount.

Estimating logistic regression coefficients via MLE

In logistic regression, we assume that

- $p_i = \Pr(Y_i = 1 \mid X_i = x_i) = \sigma(\beta_0 + \beta_1 x_i)$ for each training point x_i
- $Y_i \sim \mathrm{Bernoulli}(p_i), \qquad p_i = \sigma(\beta_0 + \beta_1 x_i)$

Assuming the n observations are independent, the conditional likelihood of the observed labels under the logistic model with parameters (β_0, β_1) is

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} \Pr(Y_i = y_i \mid X_i = x_i) = \prod_{i=1}^{n} p_i^{y_i}(1 - p_i)^{1-y_i}.$$

Maximum likelihood estimation (MLE) chooses $\hat{\beta}_0, \hat{\beta}_1$ that maximizes L (the parameters for which the given data are most likely)

- It reflects the Bernoulli nature of the response
- Unlike simple linear regression, there is no closed-form formula; software finds the MLE numerically

Why MLE instead of least squares?

Least squares criterion:

$$\min_{\beta_0,\beta_1} \sum_{i=1}^{n} (y_i - p_i)^2$$

- Fitting by squared-distance minimization
- Natural for quantitative responses with roughly constant noise
- Not the likelihood for binary $Y_i \in \{0, 1\}$

MLE criterion:

$$\max_{\beta_0,\beta_1} \prod_{i=1}^{n} p_i^{y_i}(1 - p_i)^{1-y_i}$$

$$\min_{\beta_0,\beta_1} -\sum_{i=1}^{n} \{y_i \log p_i + (1 - y_i) \log(1 - p_i)\}$$

- Assumes $Y_i \mid X_i = x_i \sim \mathrm{Bernoulli}(p_i)$
- Natural for probability estimation and inference

Practical intuition: Under the Bernoulli model, MLE is the natural fitting criterion and heavily penalizes confident wrong probability predictions

Observed $y = 1$	$p = 0.9$	$p = 0.5$	$p = 0.1$	$p = 0.01$	$p = 0.001$
$(y - p)^2$	0.01	0.25	0.81	0.9801	0.998
$-\log(p)$	0.105	0.693	2.303	4.605	6.908

Wrap-up

- Classification predicts a categorical response, not a quantitative one
- Linear regression is usually a poor choice for classification because fitted values can fall outside $[0, 1]$ and class coding can be artificial
- Logistic regression models the log-odds linearly:

$$\log\left(\frac{p(x)}{1 - p(x)}\right) = \beta_0 + \beta_1 x$$

- The logistic function converts the linear predictor into a valid probability $p(x) \in (0, 1)$, and class predictions come from thresholding estimated probabilities
- Logistic regression is typically fit by maximum likelihood estimation (MLE) under a Bernoulli model, not by least squares

Next lecture: Extensions of logistic regression, Decision boundaries, Classification errors

References

Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.
An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.
Springer, New York, NY, 2nd edition, 2021.